🏠 SMART HOME LIGHTING SYSTEM

1.Title

Design of Smart Home Lighting Automation System

2.Aim

To design a automations smart home lighting system.

3. Components Required

Arduino Uno

PIR Sensor

LDR ,LED

Power Supply (5V)

Breadboard

Connecting wires

4.Working principal

The automatic lighting system for the smart home works on the principles of light detection and movement detection. The level of light intensity in the surrounding area is detected using an LDR (light-dependent resistor). The resistance of an LDR varies in inverse proportion to the intensity of light. Therefore, when there is low light, the resistance rises, and this variation is detected via an analog input port on the Arduino board.

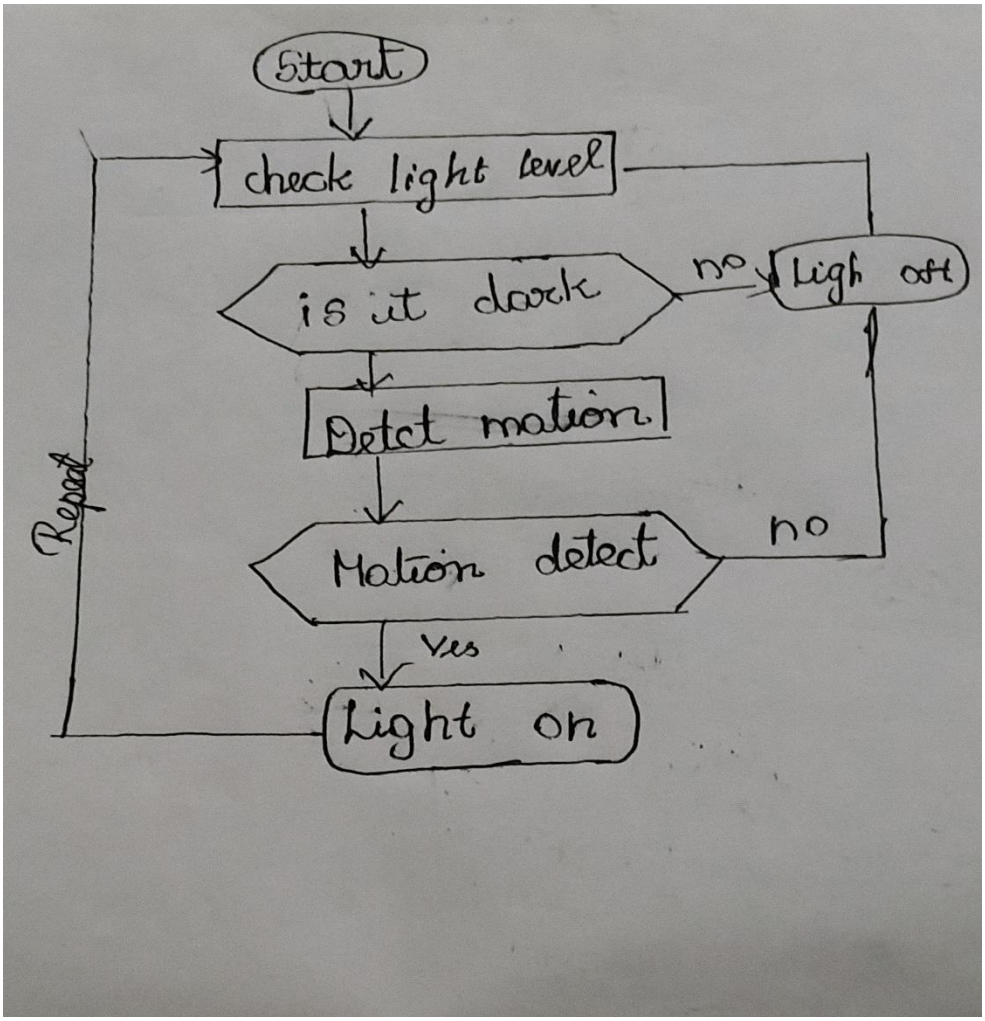
Concurrently, the presence of people in the room is detected using a passive infrared (PIR) sensor. This sensor detects changes in infrared radiation caused by the movement of any object in its field of view and produces a digital output signal.

Both sensors' inputs are monitored constantly by the Arduino board, which evaluates these inputs based on programmed logic. It is decided beforehand what level of light will indicate a bright or dark environment. If the light intensity is lower than the preset value and there is motion, then the control signal is sent from the Arduino board to the relay circuit.

A relay is basically a switching mechanism that turns the lighting fixture on when the Arduino board activates the relay circuit

5.Flow chart

t



6.Advantage

Saves electrical energy

Automatic operation

Reduces human effort

Increases safety and convenience

7 Applications

. Smart homes

Street lighting systems

Offices and buildings

Security lighting

8.Conclusion

. The smart home lighting system was successfully designed using an LDR, PIR sensor, and Arduino. The system automatically controls lighting based on ambient light and human presence, ensuring efficient energy usage. It reduces manual effort and improves convenience by turning lights ON only when required. This project demonstrates a simple, cost-effective, and reliable solution for modern home automation systems.